

A-Identity Trust Oracle

The identity and reputation oracle for the agent economy.

OpenAPI 3.1.0 | version 1.1.0 | <https://a-identity.xyz> | MIT
Generated from the live specification at <https://a-identity-asp.onrender.com/openapi.json>

Overview

Before one AI agent pays another, it calls A-Identity to verify the counterparty. Seven x402 tools (six paid, one free) over a live on-chain engine: ERC-8004 identity + KYA (Know-Your-Agent), a deterministic 0-1000 reputation score computed from real on-chain settlements, a pre-transaction ALLOW / WARN / DENY risk verdict, a two-sided counterparty check that catches same-operator self-dealing, and a guardrail policy check. Every number is reproducible and every settlement is verifiable on-chain at /proof. Paid tools use the x402 protocol in USDC: an unpaid request returns HTTP 402 with the payment requirements; after payment the request is replayed and served.

Two settlement rails are live. The tool endpoints settle per call on X Layer (eip155:196). A second rail settles on Circle Arc testnet (eip155:5042002) through Circle Gateway's batched nanopayments, where the buyer signs an EIP-3009 authorization off-chain for zero gas and thousands of sub-cent authorizations net into one on-chain transfer. On Arc the gas token is USDC itself, so an agent needs no second asset to transact.

Endpoints at a glance

ENDPOINT	PRICE	WHAT IT ANSWERS
POST /tools/verify_agent	\$0.001	Verify an AI agent's identity (ERC-8004 + KYA)
POST /tools/reputation_score	\$0.002	On-chain 0-1000 reputation score
POST /tools/risk_check	\$0.005	Pre-transaction ALLOW / WARN / DENY risk verdict
POST /tools/agent_passport	\$0.01	Full agent passport (identity + reputation + KYA + risk)
POST /tools/counterparty_check	\$0.008	Two-sided counterparty check before an agent-to-agent deal
POST /tools/guardrail_check	\$0.005	Policy guardrail check for a proposed action
GET /api/x402/nano/data	\$0.001	Live Arc chain data, paid by gasless USDC nanopayment on Arc
GET /health	free	Liveness + service card (free)
GET /.well-known/agent.json	free	Inspectable service manifest (free)
GET /proof	free	Public, verifiable proof of real on-chain settlements (free)
GET /methodology	free	Deterministic scoring methodology (free)
POST /tools/trust_preview	free	Free coarse trust band (rate limited)
GET /stats	free	Live settlement stats
GET /openapi.json	free	This document

Settlement rails

X Layer mainnet (per-call)

Network	eip155:196
Scheme	exact
Asset	USD₪0
Applies to	/tools/*

Every paid tool call settles as its own on-chain transfer. The full settlement list is public at /proof.

Circle Arc testnet (nanopayment)

Network	eip155:5042002
Scheme	exact
Asset	USDC 0x3600
Facilitator	Circle Gateway (GatewayWalletBatched)
Verifying contract	0x0077777d7eba4688bdef3e311b846f25870a19b9
Applies to	/api/x402/nano/data

Gasless for the buyer: an EIP-3009 authorization is signed off-chain and Circle Gateway batches the on-chain settlement.

Paid endpoints

POST

/tools/verify_agent

\$0.001

Verify an AI agent's identity (ERC-8004 + KYA)

Returns the agent's on-chain ERC-8004 registration and KYA (wallet-control) status. Price: \$0.001 (x402, USDC).

FIELD	TYPE	DESCRIPTION
agentId	string required	Platform id, ERC-8004 token id (e.g. "#849980"), or the owner 0x address. e.g. #849980

STATUS	MEANING
200	Identity + KYA result
400	Missing or invalid agentId.
402	x402 Payment Required. The response body carries the payment requirements; pay in USDC and replay the request.

POST /tools/reputation_score

\$0.002

On-chain 0-1000 reputation score

A deterministic 0-1000 reputation score computed from real on-chain settlements, KYA, and tenure. Price: \$0.002 (x402, USDC).

FIELD	TYPE	DESCRIPTION
agentId	string required	Platform id, ERC-8004 token id (e.g. "#849980"), or the owner 0x address. e.g. #849980
STATUS MEANING		
200	Reputation result	
400	Missing or invalid agentId.	
402	x402 Payment Required. The response body carries the payment requirements; pay in USDC and replay the request.	

POST /tools/risk_check

\$0.005

Pre-transaction ALLOW / WARN / DENY risk verdict

A single ALLOW / WARN / DENY decision with reasons for a counterparty, before you pay it. Optionally takes a txContext to size the risk to the payment. Price: \$0.005 (x402, USDC).

FIELD	TYPE	DESCRIPTION
agentId	string required	Platform id, ERC-8004 token id (e.g. "#849980"), or the owner 0x address. e.g. #849980
txContext	object optional	Optional context to size the risk to the intended payment.
STATUS MEANING		
200	Risk verdict	
400	Missing or invalid agentId.	
402	x402 Payment Required. The response body carries the payment requirements; pay in USDC and replay the request.	

POST /tools/agent_passport

\$0.01

Full agent passport (identity + reputation + KYA + risk)

The full passport in one call: ERC-8004 identity, KYA, reputation, and risk. Price: \$0.01 (x402, USDC).

FIELD	TYPE	DESCRIPTION
agentId	string required	Platform id, ERC-8004 token id (e.g. "#849980"), or the owner 0x address. e.g. #849980
STATUS MEANING		
200	Agent passport	
400	Missing or invalid agentId.	
402	x402 Payment Required. The response body carries the payment requirements; pay in USDC and replay the request.	

POST /tools/counterparty_check

\$0.008

Two-sided counterparty check before an agent-to-agent deal

Checks BOTH sides of a proposed deal and flags same-operator self-dealing, which a one-sided scan cannot see. Price: \$0.008 (x402, USDC).

FIELD	TYPE	DESCRIPTION
agentId	string required	Platform id, ERC-8004 token id (e.g. "#849980"), or the owner 0x address. e.g. #849980
txContext	object optional	Optional context to size the risk to the intended payment.
STATUS MEANING		
200	Two-sided verdict	
400	Missing or invalid agentId.	
402	x402 Payment Required. The response body carries the payment requirements; pay in USDC and replay the request.	

POST /tools/guardrail_check

\$0.005

Policy guardrail check for a proposed action

Runs the live policy engine over a proposed action and answers allow / ask a human / block, with the rule that decided it. Price: \$0.005 (x402, USDC).

FIELD	TYPE	DESCRIPTION
agentId	string required	Platform id, ERC-8004 token id (e.g. "#849980"), or the owner 0x address. e.g. #849980
txContext	object optional	Optional context to size the risk to the intended payment.
STATUS MEANING		
200	Guardrail decision	
400	Missing or invalid agentId.	
402	x402 Payment Required. The response body carries the payment requirements; pay in USDC and replay the request.	

GET /api/x402/nano/data

\$0.001

Live Arc chain data, paid by gasless USDC nanopayment on Arc

The Arc settlement rail. An unpaid GET returns an x402 v2 challenge whose accepts[] names Circle Arc testnet (eip155:5042002), native USDC, and Circle Gateway's GatewayWalletBatched domain. The buyer signs an EIP-3009 authorization off-chain, so the call costs the buyer no gas, and Gateway batches the on-chain settlement. Price: \$0.001 (1000 USDC units).

Host: <https://a-identity-backend.onrender.com>

Settles on: eip155:5042002

STATUS	MEANING
200	The paid resource, plus the Gateway settlement in the PAYMENT-RESPONSE header
402	x402 Payment Required. The response body carries the payment requirements; pay in USDC and replay the request.

Free endpoints

GET /health

free

Liveness + service card (free)

STATUS	MEANING
200	Service card (tools, prices, payment info)

GET /.well-known/agent.json

free

Inspectable service manifest (free)

STATUS	MEANING
200	The Agent Marketplace service manifest

GET /proof

free

Public, verifiable proof of real on-chain settlements (free)

STATUS	MEANING
200	Proof (HTML for a browser, JSON for an API caller)

GET /methodology

free

Deterministic scoring methodology (free)

STATUS	MEANING
200	Methodology

POST /tools/trust_preview

free

Free coarse trust band (rate limited)

A free, coarse trust band plus revoked and Sybil flags. No payment and no key; rate limited to 20 calls per hour per IP. The on-ramp before a paid call.

FIELD	TYPE	DESCRIPTION
agentId	string required	Platform id, ERC-8004 token id (e.g. "#849980"), or the owner 0x address. e.g. #849980

STATUS	MEANING
200	Coarse trust band
400	Missing or invalid agentId.

GET /stats

free

Live settlement stats

Live balance of the payout wallet and settlement totals.

STATUS	MEANING
200	Live stats

GET /openapi.json

free

This document

The OpenAPI spec for this service, served so agents and the Circle Agent Marketplace can read its inputs and outputs.

STATUS	MEANING
200	OpenAPI 3.1 document

Every paid endpoint answers an unpaid request with HTTP 402 and a machine-readable payment challenge carrying the input schema and a worked example, then serves the resource on a paid retry. Settlements are published at /proof and the scoring method at /methodology.